Supplementary Information

Consistency, not plausibility: why tool-using science agents relay database errors they detect

George Xu

Supplementary Note 1: the first deployment run and its audit

The first full run of the deployment experiment began on 25 October 2026 under the author's own macOS account. Biomni's system prompt lists its data-lake directory, and that directory sat inside the study's repository, so the path shown to the model contained the strings `bioagent-plausibility` and `biomni_spike`. Biomni's code executor was not sandboxed: code the agent wrote could read the author's filesystem and reach the internet, and the study's repository was public.

After the run, every transcript was audited with three regular expressions, recorded in `results/analysis/audit_biomni_environment.py`: *awareness* (phrases inferring a plausibility benchmark, a test, or an evaluation of the agent), *repository pattern* (code executions referencing the author's home directory outside the data lake, the study's repository files, `git` commands or process listings) and *metadata vocabulary* (terms from the study's task files, such as `corrupted_value` or "answer key"). DeepSeek V4.1 Flash matched none of the first two in any run and the third in one run (u015), which caught the corruption regardless. Gemini 3.8 Flash matched awareness in 44 runs, the repository pattern in 98 and the vocabulary in 2. Awareness changed behaviour: Flash's aware runs relayed plausible values in 1 of 23 delivered runs, against 68 of 265 for the others, so the first run's Flash catch rate was inflated. The external files Flash fetched were Biomni's own repository (its evaluation task list) and a third-party benchmark (Bio-Bench); no transcript shows the study's files being read.

The corrupted conditions were rerun on 3 November 2026 under a second macOS user with no read access to the author's home directory, with the data lake at a neutral path (`/Users/Shared/agentrun/workspace`). The agent's state file carried only the task identifier and a slot number; the mapping from slot to condition (`harness/rerun_schedule.json`, commit 0019d72) was read by the proxy and never shown to the agent. The clean runs were retained from the first run. In the rerun no transcript named the study, its repository or its paper. DeepSeek matched no pattern in any run. Flash matched awareness in 16 runs, each inferred from the task and the anomaly themselves rather than from the environment, the repository pattern in 17 runs of directory and process probing that found nothing, and the vocabulary in 2; the external files it fetched were again Biomni's own repository and Bio-Bench. Supplementary Table 1 gives both runs side by side; the main text reports the rerun throughout.

Supplementary Note 2: registration and run timestamps

Each registration was committed before the runs it concerns; the intervals were: new-field experiment, commit 2059c3a, three minutes before the first run; its agreement follow-up, 217ea9d, one minute; historical-error experiment, ac99248, 24 seconds; robustness addendum, 6830c57 and

76e56b6, 13 minutes and one minute before the respective runs; deployment hypotheses, OSF draft four minutes before the first full run, cited by commit a2dc4a4 two minutes before the runs; memory probe, 74415d3, and the prompt, repetition and withholding experiments, 499d317, each committed before the runs they concern on 3 November 2026. Commit times are recorded in the repository history.

Supplementary Table 1: Deployment experiment, first run (October 2026) and rerun (3 November 2026). Relay is relayed/delivered runs; the odds ratio compares relay of plausible values on exact-route fields with other fields (registered test, task-clustered logistic regression, 95% interval). Audit columns count runs matching each pattern.

Model	Run	Impossible relayed	Plausible relayed	Odds ratio	Aware	Repository	Vocabulary
DeepSeek V4.1 Flash	first	5/287 (2%)	80/285 (28%)	5.7 (3.1–10.4)	0	0	1
DeepSeek V4.1 Flash	rerun	4/293 (1%)	50/290 (17%)	17.2 (6.0–49.7)	0	0	0
Gemini 3.8 Flash	first	6/286 (2%)	69/288 (24%)	3.6 (2.0–6.5)	44	98	2
Gemini 3.8 Flash	rerun	4/283 (1%)	57/276 (21%)	6.9 (3.3–14.3)	16	17	2

Supplementary Table 2: Prompt variants: wrong value relayed (relayed/parsed runs) under the registered default prompt and three variants, on (a) protein tasks whose full record contradicts the value, (b) the chemistry agreement cell (three samples) and (c) coherent protein errors. The last column gives, for the two-fields variant, runs in which the stated database value matched the record (scored on (a) and (b)).

Cell	Model	Default	No tool instruction	Database has errors	Two fields	Stated value = record
(a)	Opus 5	9/49 (18%)	4/49 (8%)	0/38 (0%)	5/47 (11%)	12/47 (26%)
(a)	Gemini 4 Argon	7/50 (14%)	6/50 (12%)	1/50 (2%)	0/50 (0%)	0/50 (0%)
(a)	DeepSeek V4.1 Flash	50/50 (100%)	42/49 (86%)	23/42 (55%)	34/47 (72%)	13/47 (28%)
(b)	Opus 5	163/180 (91%)	85/180 (47%)	0/180 (0%)	0/180 (0%)	180/180 (100%)
(b)	Gemini 4 Argon	144/180 (80%)	92/180 (51%)	0/180 (0%)	0/180 (0%)	126/180 (70%)
(b)	Gemini 3.8 Flash	159/174 (91%)	58/171 (34%)	0/180 (0%)	6/163 (4%)	103/163 (63%)
(b)	DeepSeek V4.1 Flash	152/180 (84%)	140/180 (78%)	2/180 (1%)	19/180 (11%)	171/180 (95%)
(c)	Opus 5	49/50 (98%)	48/50 (96%)	50/50 (100%)	26/50 (52%)	–
(c)	Gemini 4 Argon	48/50 (96%)	47/50 (94%)	50/50 (100%)	0/50 (0%)	–
(c)	DeepSeek V4.1 Flash	47/50 (94%)	46/50 (92%)	49/49 (100%)	45/50 (90%)	–

Supplementary Table 3: Repetition versus consistency: wrong molecular weight relayed (relayed/parsed runs; 60 compounds, three samples, full record). R0, weight alone; R1, a verbatim repeat field scaled with the weight; R2, the repeat left correct; B, the heavy-atom count scaled with the weight. On uncorrupted records with the repeat field present, every model answered correctly in 60 of 60 runs. Pooled discordant pairs for RC1 (R0 relayed and R1 not, against the reverse): 198 against 84, two-sided McNemar $p < 10^{-10}$; for RC3 (R2 against R0), $p < 10^{-10}$.

Model	R0 weight alone	R1 repeat agrees	R2 repeat disagrees	B derived field agrees
Opus 5	102/180 (57%)	66/180 (37%)	0/180 (0%)	113/180 (63%)
Gemini 4 Argon	117/180 (65%)	67/180 (37%)	3/180 (2%)	130/180 (72%)
Gemini 3.8 Flash	152/177 (86%)	135/178 (76%)	1/180 (1%)	154/175 (88%)
DeepSeek V4.1 Flash	115/180 (64%)	105/179 (59%)	5/180 (3%)	128/180 (71%)
Pooled	486/717 (68%)	373/717 (52%)	9/720 (1%)	525/715 (73%)

Supplementary Table 4: Memory probe on the 79 historical proteins. Memory state is the model's answer, without tools, for the protein's length: current (within one residue of today's length), superseded (within one of the superseded length), either (both, where they differ by at most two) or other. Relay of the superseded value is given as relayed/runs by memory state, without a second source (full record, one sample) and with the second source shown (annotate arm, three samples); p is the registered one-sided Fisher test of superseded-memory against current-memory relay with the second source.

Model	Memory state	Proteins	No second source	Second source shown	p
Opus 5	current	48	47/48	2/144	
	superseded	7	7/7	13/21	$< 10^{-10}$
	either	12	12/12	0/36	
	other	12	12/12	9/36	
Gemini 4 Argon	current	53	48/53	4/159	
	superseded	6	6/6	0/17	1
	either	12	11/12	1/36	
	other	8	6/8	0/24	
DeepSeek V4.1 Flash	current	50	50/50	38/150	
	superseded	6	6/6	17/18	1×10^{-8}
	either	11	11/11	7/33	
	other	12	11/12	18/36	

Supplementary Table 5: Withholding harness (full record, 100 tasks per domain and condition, one sample): corrupted numeric fields and identifiers relayed (relayed/parsed runs), runs without a parseable answer, and runs answered with a value that was neither the true nor the corrupted one. The last column gives identifier relay in the annotate arm of the benchmark (summary record) for comparison; the arms differ in record format.

Model	Domain	Condition	Numeric relayed	Identifiers relayed	Unparsed	Wrong	Annotate arm identifiers
Opus 5	proteins	impossible	0/46	0/50	4	0	0/100
		plausible	0/45	0/50	5	0	0/100
	molecules	impossible	0/75	5/25	0	0	21/25
		plausible	0/75	3/25	0	0	10/25
Gemini 4 Argon	proteins	impossible	0/49	0/50	1	0	0/100
		plausible	0/48	0/50	2	0	0/100
	molecules	impossible	0/75	4/25	0	0	19/25
		plausible	0/75	1/25	0	0	18/25
DeepSeek V4.1 Flash	proteins	impossible	0/34	4/50	16	5	1/100
		plausible	0/37	1/50	13	3	1/100
	molecules	impossible	0/75	12/25	0	0	22/24
		plausible	0/75	11/25	0	0	21/24

Supplementary Tables 6–13: counts behind Figures 1, 2, 4 and 6 and Extended Data Figures 1 and 2

Supplementary Table 6: Deployment rerun (Fig. 1b, c): corrupted values relayed by field (relayed/delivered runs) and, among relayed runs, how many raised a concern.

Model	Condition	Field	Relayed	Flagged among relayed
DeepSeek V4.1 Flash	impossible	gene symbol	2/50 (4%)	1/2
DeepSeek V4.1 Flash	impossible	h bond donors	0/24 (0%)	–
DeepSeek V4.1 Flash	impossible	heavy atom count	0/25 (0%)	–
DeepSeek V4.1 Flash	impossible	length	0/46 (0%)	–
DeepSeek V4.1 Flash	impossible	mass	0/50 (0%)	–
DeepSeek V4.1 Flash	impossible	molecular weight	0/23 (0%)	–
DeepSeek V4.1 Flash	impossible	name	2/25 (8%)	2/2
DeepSeek V4.1 Flash	impossible	organism	0/50 (0%)	–
DeepSeek V4.1 Flash	plausible	gene symbol	0/49 (0%)	–
DeepSeek V4.1 Flash	plausible	h bond donors	4/25 (16%)	4/4
DeepSeek V4.1 Flash	plausible	heavy atom count	0/25 (0%)	–
DeepSeek V4.1 Flash	plausible	length	23/44 (52%)	6/23
DeepSeek V4.1 Flash	plausible	mass	18/49 (37%)	4/18
DeepSeek V4.1 Flash	plausible	molecular weight	5/24 (21%)	0/5
DeepSeek V4.1 Flash	plausible	name	0/24 (0%)	–
DeepSeek V4.1 Flash	plausible	organism	0/50 (0%)	–
Gemini 3.8 Flash	impossible	gene symbol	0/50 (0%)	–
Gemini 3.8 Flash	impossible	h bond donors	0/25 (0%)	–
Gemini 3.8 Flash	impossible	heavy atom count	0/22 (0%)	–
Gemini 3.8 Flash	impossible	length	1/48 (2%)	0/1
Gemini 3.8 Flash	impossible	mass	1/48 (2%)	1/1
Gemini 3.8 Flash	impossible	molecular weight	0/21 (0%)	–
Gemini 3.8 Flash	impossible	name	1/23 (4%)	0/1
Gemini 3.8 Flash	impossible	organism	1/46 (2%)	1/1
Gemini 3.8 Flash	plausible	gene symbol	1/49 (2%)	1/1
Gemini 3.8 Flash	plausible	h bond donors	5/23 (22%)	4/5
Gemini 3.8 Flash	plausible	heavy atom count	0/23 (0%)	–
Gemini 3.8 Flash	plausible	length	16/44 (36%)	7/16
Gemini 3.8 Flash	plausible	mass	30/48 (62%)	9/30
Gemini 3.8 Flash	plausible	molecular weight	1/21 (5%)	1/1
Gemini 3.8 Flash	plausible	name	0/20 (0%)	–
Gemini 3.8 Flash	plausible	organism	4/48 (8%)	3/4

Supplementary Table 7: Benchmark (Extended Data Fig. 1): corrupted values relayed, runs raising a concern, and relayed runs that raised a concern (summary record, registered prompt, one sample).

Model	Domain	Condition	Relayed	Flagged	Flagged among relayed
Opus 5	proteins	impossible	64/200 (32%)	200/200 (100%)	64/64
Opus 5	proteins	plausible	150/200 (75%)	118/200 (59%)	68/150
Opus 5	molecules	impossible	46/100 (46%)	100/100 (100%)	46/46
Opus 5	molecules	plausible	98/100 (98%)	97/100 (97%)	95/98
Gemini 4 Argon	proteins	impossible	70/200 (35%)	200/200 (100%)	70/70
Gemini 4 Argon	proteins	plausible	147/199 (74%)	189/199 (95%)	137/147
Gemini 4 Argon	molecules	impossible	62/100 (62%)	91/100 (91%)	53/62
Gemini 4 Argon	molecules	plausible	98/100 (98%)	90/100 (90%)	88/98
DeepSeek V4.1 Flash	proteins	impossible	88/200 (44%)	200/200 (100%)	88/88
DeepSeek V4.1 Flash	proteins	plausible	154/200 (77%)	92/200 (46%)	46/154
DeepSeek V4.1 Flash	molecules	impossible	38/100 (38%)	97/100 (97%)	35/38
DeepSeek V4.1 Flash	molecules	plausible	95/99 (96%)	71/99 (72%)	67/95
GPT-6.1 Sol	proteins	impossible	133/200 (66%)	200/200 (100%)	133/133
GPT-6.1 Sol	proteins	plausible	150/200 (75%)	105/200 (52%)	55/150
GPT-6.1 Sol	molecules	impossible	67/100 (67%)	78/100 (78%)	45/67
GPT-6.1 Sol	molecules	plausible	93/100 (93%)	71/100 (71%)	64/93
GPT-6 Luna	proteins	impossible	146/200 (73%)	163/200 (82%)	109/146
GPT-6 Luna	proteins	plausible	171/200 (86%)	34/200 (17%)	6/171
GPT-6 Luna	molecules	impossible	68/100 (68%)	75/100 (75%)	43/68
GPT-6 Luna	molecules	plausible	100/100 (100%)	16/100 (16%)	16/100
Kimi K3	proteins	impossible	149/200 (74%)	199/200 (100%)	148/149
Kimi K3	proteins	plausible	184/200 (92%)	58/200 (29%)	42/184
Kimi K3	molecules	impossible	63/100 (63%)	88/100 (88%)	51/63
Kimi K3	molecules	plausible	99/100 (99%)	60/100 (60%)	59/99
Sonnet 5	proteins	impossible	151/200 (76%)	200/200 (100%)	151/151
Sonnet 5	proteins	plausible	185/200 (92%)	63/200 (32%)	48/185
Sonnet 5	molecules	impossible	73/100 (73%)	90/100 (90%)	63/73
Sonnet 5	molecules	plausible	100/100 (100%)	50/100 (50%)	50/100
Sonnet 4.6	proteins	impossible	166/200 (83%)	199/200 (100%)	165/166
Sonnet 4.6	proteins	plausible	199/200 (100%)	50/200 (25%)	49/199
Sonnet 4.6	molecules	impossible	83/100 (83%)	84/100 (84%)	67/83
Sonnet 4.6	molecules	plausible	100/100 (100%)	38/100 (38%)	38/100
Gemini 3.8 Flash	proteins	impossible	167/198 (84%)	195/198 (98%)	164/167
Gemini 3.8 Flash	proteins	plausible	155/200 (78%)	64/200 (32%)	19/155
Gemini 3.8 Flash	molecules	impossible	62/99 (63%)	77/99 (78%)	40/62
Gemini 3.8 Flash	molecules	plausible	100/100 (100%)	56/100 (56%)	56/100
Haiku 4.5	proteins	impossible	186/200 (93%)	152/200 (76%)	138/186
Haiku 4.5	proteins	plausible	200/200 (100%)	15/200 (8%)	15/200
Haiku 4.5	molecules	impossible	91/100 (91%)	75/100 (75%)	66/91
Haiku 4.5	molecules	plausible	100/100 (100%)	3/100 (3%)	3/100

Supplementary Table 8: Override boundaries (Extended Data Fig. 2), value decreased: corrupted value relayed (relayed/parsed runs) at each corruption factor, full record, paired field scaled alike.

Model	Domain	×0.9	×0.8	×0.7	×0.5	×0.3	×0.1	×0.02	×0.005
Opus 5	proteins	18/40	8/39	1/40	0/39	0/39	0/39	0/38	0/33
Opus 5	molecules	16/40	14/40	9/40	4/40	0/40	1/40	0/40	0/40
Sonnet 5	proteins	37/40	35/40	28/40	15/38	5/40	0/40	0/40	0/38
Sonnet 5	molecules	40/40	40/40	33/40	17/40	3/40	0/40	0/40	0/40
Gemini 4 Argon	proteins	23/40	16/40	9/40	1/40	0/40	0/40	0/40	0/40
Gemini 4 Argon	molecules	28/40	24/40	22/40	11/40	3/40	0/40	0/40	0/40
Gemini 3.8 Flash	proteins	21/39	17/39	12/38	5/37	2/40	0/40	0/40	0/40
Gemini 3.8 Flash	molecules	29/39	22/35	15/39	15/40	7/39	2/40	5/40	4/40
DeepSeek V4.1 Flash	proteins	38/40	33/39	33/39	21/39	8/36	0/35	1/30	0/30
DeepSeek V4.1 Flash	molecules	22/40	11/40	10/40	6/40	1/39	0/40	0/40	0/40

Supplementary Table 9: Override boundaries (Extended Data Fig. 2), value increased: corrupted value relayed (relayed/parsed runs) at each corruption factor, full record, paired field scaled alike.

Model	Domain	×1.1	×1.25	×1.5	×2	×3	×10	×50	×200
Opus 5	proteins	25/40	6/38	4/40	0/40	0/40	0/40	0/40	0/40
Opus 5	molecules	19/40	10/40	7/40	4/40	4/40	0/40	0/40	0/40
Sonnet 5	proteins	40/40	37/40	32/40	24/40	12/39	0/40	0/38	0/40
Sonnet 5	molecules	40/40	40/40	40/40	36/40	30/40	4/40	2/40	0/40
Gemini 4 Argon	proteins	15/40	11/40	4/40	0/40	0/40	0/40	0/40	0/40
Gemini 4 Argon	molecules	22/40	19/40	11/40	13/40	5/40	0/40	0/40	0/40
Gemini 3.8 Flash	proteins	26/40	17/39	6/39	1/40	0/40	0/40	0/40	0/40
Gemini 3.8 Flash	molecules	21/40	21/39	17/39	14/40	18/38	0/40	4/39	0/40
DeepSeek V4.1 Flash	proteins	40/40	31/40	24/39	13/38	3/34	0/38	1/33	0/38
DeepSeek V4.1 Flash	molecules	30/40	22/40	19/40	13/40	5/40	0/40	0/40	0/40

Supplementary Table 10: New chemistry fields with the full record (Fig. 4a): runs raising a concern and corrupted values relayed, by field and condition.

Model	Field	Condition	Flagged	Relayed
Opus 5	complexity	clean	1/60 (2%)	–
Opus 5	complexity	plausible	1/60 (2%)	60/60 (100%)
Opus 5	complexity	impossible	60/60 (100%)	60/60 (100%)
Opus 5	monoisotopic mass	clean	2/60 (3%)	–
Opus 5	monoisotopic mass	plausible	52/60 (87%)	57/60 (95%)
Opus 5	monoisotopic mass	impossible	60/60 (100%)	0/60 (0%)
Opus 5	tpsa	clean	1/60 (2%)	–
Opus 5	tpsa	plausible	10/60 (17%)	60/60 (100%)
Opus 5	tpsa	impossible	60/60 (100%)	19/60 (32%)
Sonnet 5	complexity	clean	3/60 (5%)	–
Sonnet 5	complexity	plausible	2/60 (3%)	60/60 (100%)
Sonnet 5	complexity	impossible	60/60 (100%)	60/60 (100%)
Sonnet 5	monoisotopic mass	clean	6/60 (10%)	–
Sonnet 5	monoisotopic mass	plausible	36/60 (60%)	59/60 (98%)
Sonnet 5	monoisotopic mass	impossible	60/60 (100%)	2/60 (3%)
Sonnet 5	tpsa	clean	0/60 (0%)	–
Sonnet 5	tpsa	plausible	0/60 (0%)	60/60 (100%)
Sonnet 5	tpsa	impossible	60/60 (100%)	41/60 (68%)
Gemini 4 Argon	complexity	clean	0/60 (0%)	–
Gemini 4 Argon	complexity	plausible	1/60 (2%)	60/60 (100%)
Gemini 4 Argon	complexity	impossible	60/60 (100%)	51/60 (85%)
Gemini 4 Argon	monoisotopic mass	clean	3/60 (5%)	–
Gemini 4 Argon	monoisotopic mass	plausible	60/60 (100%)	58/60 (97%)
Gemini 4 Argon	monoisotopic mass	impossible	60/60 (100%)	0/60 (0%)
Gemini 4 Argon	tpsa	clean	1/60 (2%)	–
Gemini 4 Argon	tpsa	plausible	47/60 (78%)	60/60 (100%)
Gemini 4 Argon	tpsa	impossible	60/60 (100%)	12/60 (20%)
Gemini 3.8 Flash	complexity	clean	0/60 (0%)	–
Gemini 3.8 Flash	complexity	plausible	0/60 (0%)	60/60 (100%)
Gemini 3.8 Flash	complexity	impossible	60/60 (100%)	57/60 (95%)
Gemini 3.8 Flash	monoisotopic mass	clean	0/60 (0%)	–
Gemini 3.8 Flash	monoisotopic mass	plausible	58/60 (97%)	60/60 (100%)
Gemini 3.8 Flash	monoisotopic mass	impossible	58/58 (100%)	1/58 (2%)
Gemini 3.8 Flash	tpsa	clean	0/60 (0%)	–
Gemini 3.8 Flash	tpsa	plausible	4/60 (7%)	59/60 (98%)
Gemini 3.8 Flash	tpsa	impossible	56/56 (100%)	30/56 (54%)
DeepSeek V4.1 Flash	complexity	clean	6/60 (10%)	–
DeepSeek V4.1 Flash	complexity	plausible	5/60 (8%)	60/60 (100%)
DeepSeek V4.1 Flash	complexity	impossible	60/60 (100%)	60/60 (100%)
DeepSeek V4.1 Flash	monoisotopic mass	clean	2/60 (3%)	–
DeepSeek V4.1 Flash	monoisotopic mass	plausible	55/60 (92%)	56/60 (93%)
DeepSeek V4.1 Flash	monoisotopic mass	impossible	60/60 (100%)	0/60 (0%)
DeepSeek V4.1 Flash	tpsa	clean	2/60 (3%)	–
DeepSeek V4.1 Flash	tpsa	plausible	8/60 (13%)	60/60 (100%)
DeepSeek V4.1 Flash	tpsa	impossible	58/58 (100%)	13/58 (22%)

Supplementary Table 11: Agreement follow-up (Fig. 4b, c): wrong value relayed on the same 60 compounds, pooled over three samples and per sample.

Model	Arm	Pooled	Sample 1	Sample 2	Sample 3
Opus 5	mono. mass asked, two agreeing fields	175/180 (97%)	57/60 (95%)	60/60 (100%)	58/60 (97%)
Opus 5	mono. mass asked, corrupted alone	138/180 (77%)	48/60 (80%)	47/60 (78%)	43/60 (72%)
Opus 5	weight asked, one agreeing field	113/180 (63%)	31/60 (52%)	43/60 (72%)	39/60 (65%)
Opus 5	weight asked, two agreeing fields	163/180 (91%)	52/60 (87%)	58/60 (97%)	53/60 (88%)
Gemini 4 Argon	mono. mass asked, two agreeing fields	172/180 (96%)	58/60 (97%)	55/60 (92%)	59/60 (98%)
Gemini 4 Argon	mono. mass asked, corrupted alone	129/180 (72%)	43/60 (72%)	46/60 (77%)	40/60 (67%)
Gemini 4 Argon	weight asked, one agreeing field	130/180 (72%)	40/60 (67%)	44/60 (73%)	46/60 (77%)
Gemini 4 Argon	weight asked, two agreeing fields	144/180 (80%)	52/60 (87%)	44/60 (73%)	48/60 (80%)
Gemini 3.8 Flash	mono. mass asked, two agreeing fields	178/180 (99%)	60/60 (100%)	59/60 (98%)	59/60 (98%)
Gemini 3.8 Flash	mono. mass asked, corrupted alone	149/176 (85%)	50/59 (85%)	49/59 (83%)	50/58 (86%)
Gemini 3.8 Flash	weight asked, one agreeing field	154/175 (88%)	52/59 (88%)	53/59 (90%)	49/57 (86%)
Gemini 3.8 Flash	weight asked, two agreeing fields	159/174 (91%)	52/58 (90%)	54/58 (93%)	53/58 (91%)
DeepSeek V4.1 Flash	mono. mass asked, two agreeing fields	158/179 (88%)	56/60 (93%)	52/59 (88%)	50/60 (83%)
DeepSeek V4.1 Flash	mono. mass asked, corrupted alone	100/180 (56%)	36/60 (60%)	29/60 (48%)	35/60 (58%)
DeepSeek V4.1 Flash	weight asked, one agreeing field	128/180 (71%)	43/60 (72%)	41/60 (68%)	44/60 (73%)
DeepSeek V4.1 Flash	weight asked, two agreeing fields	152/180 (84%)	54/60 (90%)	48/60 (80%)	50/60 (83%)

Supplementary Table 12: Historical UniProt errors (Fig. 2): superseded value relayed with the full and summary record (one sample), full-record repeats (three samples), and with AlphaFold DB as second source (annotate arm, three samples) on covered proteins, by direction of the revision, and on the uncovered protein.

Model	Full record	Summary	Full, repeats	Second source	Superseded longer	Superseded shorter	Uncovered
Opus 5	78/79 (99%)	79/79 (100%)	90/90 (100%)	21/234 (9%)	16/63 (25%)	5/171 (3%)	3/3 (100%)
Gemini 4 Argon	71/79 (90%)	79/79 (100%)	213/236 (90%)	2/233 (1%)	1/63 (2%)	1/170 (1%)	3/3 (100%)
DeepSeek V4.1 Flash	78/79 (99%)	79/79 (100%)	234/236 (99%)	77/234 (33%)	27/63 (43%)	50/171 (29%)	3/3 (100%)

Supplementary Table 13: Presentation of the correction (Fig. 6): superseded value relayed on covered proteins under four presentations, pooled over three samples and per sample.

Model	Presentation	Pooled	Sample 1	Sample 2	Sample 3
Opus 5	annotate: conflict listed	21/234 (9%)	7/78 (9%)	7/78 (9%)	7/78 (9%)
Opus 5	replace: ambiguous labels	25/234 (11%)	7/78 (9%)	9/78 (12%)	9/78 (12%)
Opus 5	replace: original labelled	12/234 (5%)	4/78 (5%)	3/78 (4%)	5/78 (6%)
Opus 5	replace: silent	0/234 (0%)	0/78 (0%)	0/78 (0%)	0/78 (0%)
Gemini 4 Argon	annotate: conflict listed	2/233 (1%)	1/78 (1%)	0/77 (0%)	1/78 (1%)
Gemini 4 Argon	replace: ambiguous labels	17/231 (7%)	5/77 (6%)	7/77 (9%)	5/77 (6%)
Gemini 4 Argon	replace: original labelled	95/231 (41%)	37/78 (47%)	31/77 (40%)	27/76 (36%)
Gemini 4 Argon	replace: silent	0/234 (0%)	0/78 (0%)	0/78 (0%)	0/78 (0%)
DeepSeek V4.1 Flash	annotate: conflict listed	77/234 (33%)	24/78 (31%)	28/78 (36%)	25/78 (32%)
DeepSeek V4.1 Flash	replace: ambiguous labels	6/234 (3%)	1/78 (1%)	3/78 (4%)	2/78 (3%)
DeepSeek V4.1 Flash	replace: original labelled	17/233 (7%)	7/78 (9%)	5/78 (6%)	5/77 (6%)
DeepSeek V4.1 Flash	replace: silent	0/234 (0%)	0/78 (0%)	0/78 (0%)	0/78 (0%)

Supplementary Tables 14–20: experiments added in revision

Supplementary Table 14: Identity. Relay (relayed/parsed runs) on the 25 length-threshold benchmark tasks with the full record, with identity but no annotation (accession, protein name, gene, organism, length, mass, sequence), and with neither (anonymous; three samples, the first paired with the other formats); clean rows give accuracy. The last row per model gives relay of the 79 superseded sequences without a second source, with the full record and anonymously.

Model	Condition	Full record	Identity, no annotation	Neither, sample 1	Neither, 3 samples
Opus 5	clean (accuracy)	25/25 (100%)	25/25 (100%)	25/25 (100%)	75/75 (100%)
Opus 5	impossible	0/25 (0%)	0/25 (0%)	0/25 (0%)	0/75 (0%)
Opus 5	plausible	1/25 (4%)	7/25 (28%)	17/25 (68%)	50/75 (67%)
Opus 5	superseded record	78/79 (99%)	–	79/79 (100%)	–
Gemini 4 Argon	clean (accuracy)	25/25 (100%)	25/25 (100%)	25/25 (100%)	75/75 (100%)
Gemini 4 Argon	impossible	0/25 (0%)	0/25 (0%)	0/25 (0%)	0/75 (0%)
Gemini 4 Argon	plausible	0/25 (0%)	1/25 (4%)	0/25 (0%)	1/75 (1%)
Gemini 4 Argon	superseded record	71/79 (90%)	–	77/79 (97%)	–
DeepSeek V4.1 Flash	clean (accuracy)	25/25 (100%)	25/25 (100%)	25/25 (100%)	75/75 (100%)
DeepSeek V4.1 Flash	impossible	0/25 (0%)	0/24 (0%)	1/25 (4%)	2/73 (3%)
DeepSeek V4.1 Flash	plausible	22/25 (88%)	23/25 (92%)	21/25 (84%)	70/75 (93%)
DeepSeek V4.1 Flash	superseded record	78/79 (99%)	–	79/79 (100%)	–

Supplementary Table 15: Reasoning. Relay with adaptive thinking (the setting used throughout) and with thinking disabled, paired by task.

Model	Cell	Condition	Thinking adaptive	Thinking disabled
Opus 5	full record, first 50 tasks	impossible	0/50 (0%)	1/50 (2%)
Opus 5	full record, first 50 tasks	plausible	4/50 (8%)	23/50 (46%)
Sonnet 5	summary record, proteins	impossible	151/200 (76%)	139/200 (70%)
Sonnet 5	summary record, proteins	plausible	185/200 (92%)	189/200 (94%)
Sonnet 5	summary record, chemistry	impossible	73/100 (73%)	63/100 (63%)
Sonnet 5	summary record, chemistry	plausible	100/100 (100%)	100/100 (100%)

Supplementary Table 16: Request settings recorded in the traces, by model as requested (the three most frequent configurations; runs counted over the benchmark and round-2 files). Settings not listed were the provider's defaults.

Model	Settings	Runs
anthropic:claude-haiku-4-5	max_tokens=8000; model=claude-haiku-4-5; provider=anthropic; thinking=adaptive	1500
anthropic:claude-opus-5	max_tokens=8000; model=claude-opus-5; provider=anthropic; thinking=adaptive	2534
anthropic:claude-opus-5	max_tokens=8000; model=claude-opus-5; provider=anthropic; thinking=disabled	100
anthropic:claude-sonnet-4-6	max_tokens=8000; model=claude-sonnet-4-6; provider=anthropic; thinking=adaptive	1500
anthropic:claude-sonnet-5	max_tokens=8000; model=claude-sonnet-5; provider=anthropic; thinking=adaptive	1500
anthropic:claude-sonnet-5	max_tokens=8000; model=claude-sonnet-5; provider=anthropic; thinking=disabled	600
gemini:gemini-3.8-flash	max_tokens=8000; model=google/gemini-3.8-flash; provider=gemini; reasoning_effort=None; thinking=adaptive	1680
gemini:gemini-4-argon	max_tokens=8000; model=google/gemini-4-argon; provider=gemini; reasoning_effort=None; thinking=adaptive	2534
openai:gpt-6-luna	api=responses; max_tokens=8000; model=gpt-6-luna; provider=openai; reasoning_effort=None; thinking=adaptive	1500
openai:gpt-6.1-sol	api=responses; max_tokens=8000; model=gpt-6.1-sol; provider=openai; reasoning_effort=None; thinking=adaptive	1500
together:deepseek-ai/DeepSeek-V4.1-Flash-1017	max_tokens=8000; model=deepseek-ai/DeepSeek-V4.1-Flash-1017; provider=together; reasoning_effort=None; thinking=adaptive	2534
together:moonshotai/Kimi-K3	max_tokens=8000; model=moonshotai/Kimi-K3; provider=together; reasoning_effort=None; thinking=adaptive	1500

Supplementary Table 17: Prompt without the concerns field (summary record, first 100 tasks per domain). Relay under the registered prompt and without the concerns field; runs listing a concern under the registered prompt; runs voicing doubt in the reply text with and without the field, by the validated classifier; and the registered keyword detector without the field, a lower bound.

Model	Domain	Condition	Relay, registered	Relay, no field	Concerns listed	Doubt in reply, registered	Doubt in reply, no field	Keyword, no field
Opus 5	proteins	impossible	33/100 (33%)	60/100 (60%)	100/100 (100%)	99/100 (99%)	84/100 (84%)	45/100 (45%)
Opus 5	proteins	plausible	75/100 (75%)	79/100 (79%)	57/100 (57%)	52/100 (52%)	29/100 (29%)	17/100 (17%)
Opus 5	chemistry	impossible	46/100 (46%)	57/100 (57%)	100/100 (100%)	100/100 (100%)	75/100 (75%)	32/100 (32%)
Opus 5	chemistry	plausible	98/100 (98%)	100/100 (100%)	97/100 (97%)	96/100 (96%)	17/100 (17%)	0/100 (0%)
Gemini 4 Argon	proteins	impossible	37/100 (37%)	80/100 (80%)	100/100 (100%)	99/100 (99%)	36/100 (36%)	5/100 (5%)
Gemini 4 Argon	proteins	plausible	74/100 (74%)	100/100 (100%)	94/100 (94%)	85/100 (85%)	1/100 (1%)	0/100 (0%)
Gemini 4 Argon	chemistry	impossible	62/100 (62%)	39/100 (39%)	91/100 (91%)	83/100 (83%)	72/100 (72%)	31/100 (31%)
Gemini 4 Argon	chemistry	plausible	98/100 (98%)	100/100 (100%)	90/100 (90%)	81/100 (81%)	0/100 (0%)	0/100 (0%)
DeepSeek V4.1 Flash	proteins	impossible	43/100 (43%)	68/100 (68%)	100/100 (100%)	95/100 (95%)	71/100 (71%)	35/100 (35%)
DeepSeek V4.1 Flash	proteins	plausible	78/100 (78%)	83/100 (83%)	43/100 (43%)	41/100 (41%)	21/100 (21%)	8/100 (8%)
DeepSeek V4.1 Flash	chemistry	impossible	38/100 (38%)	50/100 (50%)	97/100 (97%)	87/100 (87%)	66/100 (66%)	20/100 (20%)
DeepSeek V4.1 Flash	chemistry	plausible	95/99 (96%)	100/100 (100%)	71/99 (72%)	57/99 (58%)	14/100 (14%)	2/100 (2%)

Supplementary Table 18: Familiar duplicate field. Relay of a wrong molecular weight alone and with a verbatim copy that agrees or disagrees, under an unfamiliar (molecular_weight_reported) or familiar (average_mass) field name; first sample of the earlier arms; 60 compounds.

Model	Alone	Copy agrees, unfamiliar	Copy agrees, familiar	Copy disagrees, unfamiliar	Copy disagrees, familiar	Clean accuracy, familiar copy
Opus 5	35/60 (58%)	24/60 (40%)	32/60 (53%)	0/60 (0%)	0/60 (0%)	60/60 (100%)
Gemini 4 Argon	37/60 (62%)	23/60 (38%)	32/60 (53%)	1/60 (2%)	6/60 (10%)	60/60 (100%)
Gemini 3.8 Flash	48/59 (81%)	47/59 (80%)	46/56 (82%)	0/60 (0%)	1/57 (2%)	60/60 (100%)
DeepSeek V4.1 Flash	32/60 (53%)	39/59 (66%)	40/59 (68%)	0/60 (0%)	14/60 (23%)	60/60 (100%)

Supplementary Table 19: Revision audit of the 79 superseded sequences: class, number of proteins, number with the change at the N-terminus, and relay of the superseded value pooled over Opus 5, Gemini 4 Argon and DeepSeek V4.1 Flash in four arms (second source shown: three samples).

Class	Proteins	N-terminal	Full record	Summary	Second source shown	Identity removed
documented error	13	10	33/39 (85%)	39/39 (100%)	27/117 (23%)	39/39 (100%)
flagged discrepancy	1	0	3/3 (100%)	3/3 (100%)	0/9 (0%)	3/3 (100%)
current isoform	16	10	48/48 (100%)	48/48 (100%)	25/143 (17%)	48/48 (100%)
matches a current RefSeq/Ensembl model	1	0	3/3 (100%)	3/3 (100%)	9/9 (100%)	3/3 (100%)
retired, reason not recorded	48	26	140/144 (97%)	144/144 (100%)	48/432 (11%)	142/144 (99%)

Supplementary Table 20: Runs without a parseable answer: every cell with at least two, with the reported rate (unparsed runs excluded) and the rate with unparsed runs counted against the agent (as relayed for relay, as wrong for accuracy).

Experiment	File	Condition	Parsed	Unparsed	Reported	Counted against the agent
benchmark	traces_gemini38flash	impossible	198	2	relay 167/198 (84%)	169/200 (84%)
full record	traces_deepseek_v41flash	impossible	97	3	relay 11/97 (11%)	14/100 (14%)
prompt variants	deepseek_protein_record_errors	plausible	92	8	relay 23/92 (25%)	31/100 (31%)
prompt variants	deepseek_protein_two_fields	plausible	97	3	relay 35/97 (36%)	38/100 (38%)
prompt variants	flash_agreeC_no_tool_instruction	plausible	171	9	relay 58/171 (34%)	67/180 (37%)
prompt variants	flash_agreeC_two_fields	plausible	163	17	relay 6/163 (4%)	23/180 (13%)
prompt variants	opus5_protein_record_errors	plausible	88	12	relay 0/88 (0%)	12/100 (12%)
prompt variants	opus5_protein_two_fields	plausible	97	3	relay 5/97 (5%)	8/100 (8%)
withholding	argon_protein	plausible	98	2	relay 0/98 (0%)	2/100 (2%)
withholding	deepseek_protein	impossible	84	16	relay 4/84 (5%)	20/100 (20%)
withholding	deepseek_protein	plausible	87	13	relay 1/87 (1%)	14/100 (14%)
withholding	opus5_protein	impossible	96	4	relay 0/96 (0%)	4/100 (4%)
withholding	opus5_protein	plausible	95	5	relay 0/95 (0%)	5/100 (5%)
round 2	deepseek_IS_bench	impossible	73	2	relay 2/73 (3%)	4/75 (5%)
round 2	flash_F1	plausible	56	4	relay 46/56 (82%)	50/60 (83%)
round 2	flash_F2	plausible	57	3	relay 1/57 (2%)	4/60 (7%)
superseded	argon_p5_replace	superseded	234	3	relay 20/234 (9%)	23/237 (10%)
superseded	argon_p5_replace_labeled	superseded	234	3	relay 98/234 (42%)	101/237 (43%)

Supplementary Table 21: Every registered prediction, with the commit or registration that fixed it before the runs, its test, and its outcome. Labels are those of the registrations, except that the coherent-corruption and memory-probe predictions (registered as P1–P4 and P1–P3) are prefixed C and M.

ID	Experiment	Prediction and test	Outcome	Status
DH1	Deployment (OSF jmwab)	Plausible values are relayed less often in deployment than in the benchmark (McNemar, per model)	17% and 21% against 84%; $p < 10^{-10}$ for both models	held
DH2	Deployment	Plausible corruptions of exact-route fields are relayed less often than of other fields (logistic regression, task-clustered)	Relayed more: odds ratios 17.2 (DeepSeek) and 6.9 (Flash)	failed
DH3	Deployment	Relayed values are flagged less often in deployment than in the benchmark (one-sided Fisher)	DeepSeek plausible 28% against 45% ($p = 0.016$); Flash 44% against 29%; impossible values too few ($p = 0.12$, 0.07)	partly held
H1	New fields (2059c3a)	Plausible relay rises with route class, monoisotopic mass < TPSA < complexity (one-sided Wald; Opus 5, Argon, Flash, DeepSeek)	Positive ($p = 0.007$) but negligible: 96%, 99.6%, 100%	held
H2	New fields	Monoisotopic-mass relay falls inside each model's interval from its molecular-weight curve in at least 8 of 10 cells	2 of 10 (both Sonnet 5, at ceiling); 90–100% observed, 27–59% predicted	failed
H3	New fields	The route effect is larger with the full than the summary record (interaction, one-sided)	Every summary-record run relayed	not testable
E1	Agreement follow-up (217ea9d)	Molecular weight with one agreeing field falls inside the H2 intervals in at least 8 of 10 cells	5 of 10	failed
E2	Agreement follow-up	Relay is higher with two agreeing fields than one (logistic regression, one-sided)	$p = 4 \times 10^{-7}$	held
E3	Agreement follow-up	Relay is lower when monoisotopic mass is corrupted alone	$p = 9 \times 10^{-10}$; all four models	held
C1	Coherent corruptions (28 Oct; registered as P1–P4)	Coherent plausible errors with the full record are relayed in at least 70% for all three models	70–98%	held
C2	Coherent corruptions	Coherent impossible values with the full record are relayed less than the summary-record impossible baseline, but above 0	3 of 6 comparisons as predicted	partly held
C3	Coherent corruptions	With the annotate harness and the second source, relay is at most 5% on covered entities	Argon 0%; Opus 5 and DeepSeek above 5% in some cells	partly held
C4	Coherent corruptions	An all-clear harness message raises relay of coherent impossible values from the summary record	Not raised (proteins at ceiling; chemistry lower for Opus 5)	failed
R1	Superseded sequences (ac99248)	Without a second source, relay is at least 80% in at least 5 of 6 model-by-arm cells	6 of 6; 90–100%	held
R2	Superseded sequences	With the second source, relay on covered proteins is at most 10% for each model	Opus 5 9%, Argon 1%, DeepSeek 31–36% in every sample	failed
R3	Superseded sequences	Residual relay is higher when the superseded sequence is longer than today's (one-sided Fisher)	25% against 11%, $p = 0.007$ (first sample)	held
RR1	Robustness addendum (6830c57)	Replace with ambiguous labels: relay on covered proteins at most 2% per model	11%, 7% and 3%	failed

ID	Experiment	Prediction and test	Outcome	Status
RR2	Robustness addendum	Replace relays less than annotate (one-sided Wilcoxon, pooled)	$p = 2.5 \times 10^{-4}$ pooled; heterogeneous by model	held
L1	Robustness addendum (76e56b6)	Original labelled: relay at most 2% per model	Opus 5 5%, Argon 41%, DeepSeek 7%	failed
L2	Robustness addendum	Labelled relays less than ambiguous labels (one-sided Wilcoxon)	$p = 1.0$	failed
S1	Robustness addendum	Silent correction: relay at most 2% per model	0 of 702	held
M1	Memory probe (74415d3; registered as P1–P3)	Memory holds the superseded length for at least 20% of proteins in at least one model	8–9%	failed
M2	Memory probe	With the second source shown, superseded-memory proteins are relayed more than current-memory ones (one-sided Fisher)	Opus 5 $p = 3 \times 10^{-12}$, DeepSeek $p = 1{\times}10^{-8}$; Argon 0 of 17 against 4 of 159	partly held
M3	Memory probe	Without a second source, relay is at least 90% in both memory states for every model	91–100%	held
PA1	Prompt variants (499d317)	The errors-noted prompt lowers relay on the contradicted protein cell by at least 20 points for every model	DeepSeek 100% to 55%; Opus 5 and Argon from 18% and 14% to 0% and 2%	partly held
PA2	Prompt variants	Removing the tool instruction changes relay on the protein and agreement cells by fewer than 10 points	Agreement cell fell 44, 29 and 57 points (Opus 5, Argon, Flash)	failed
PA3	Prompt variants	The two-fields format lowers relay on the protein cell by at least 20 points and states the database value faithfully in at least 90% of runs	First part DeepSeek only; value matched the record in 0–28% of runs	failed
PA4	Prompt variants	No variant lowers relay of coherent errors below 80%	Instruction variants 92–100%; two-fields 52% and 0% (Opus 5, Argon)	partly held
RC1	Repetition (499d317)	A verbatim agreeing copy raises relay	Lowered: 68% to 52% (two-sided $p < 10^{-10}$)	failed
RC2	Repetition	A derived agreeing field raises relay more than a copy, by at least 10 points pooled	73% against 52%, because the copy lowered relay	held
RC3	Repetition	A disagreeing copy lowers relay	0–3% against 57–86%	held
BH1	Withholding (499d317)	Numeric relay at most 5% for every model	0 of 709	held
BH2	Withholding	Identifier relay is unchanged from the annotate arm	Proteins 0–8% against 0–1%; compound names 4–48% against 40–92% (record formats differ)	partly held
IS1	Identity (9234013)	Superseded values are relayed in at least 90% of anonymous runs	97–100%	held
IS2	Identity	Plausible relay without identity lies within 15 points of the full record	Argon and DeepSeek within; Opus 5 68% against 4%	partly held
IS3	Identity	Impossible relay without identity is at most 20%	0–3%	held
IS4	Identity	Clean accuracy without identity is at least 90%	100%	held
IS5	Identity, addendum (c4f1f43)	Opus 5 with identity but no annotation lies within 15 points of the full or the anonymous record	28% against 4% and 68%	failed
TH1	Reasoning (9234013)	Without thinking, relay of contradicted values rises by at least 10 points	Opus 5 plausible 8% to 46% ($p = 2 \times 10^{-6}$); impossible 0% to 2%; Sonnet 5 fell	partly held

ID	Experiment	Prediction and test	Outcome	Status
TH2	Reasoning	Without thinking, plausible summary-record relay changes by fewer than 10 points	92% to 94%; 100% to 100%	held
TH3	Reasoning, addendum (cfbf0a7)	Argon at low reasoning effort relays contradicted plausible values at least 10 points more (one-sided McNemar)	6% to 14% ($p = 0.06$)	failed
NC1	No-concerns prompt (9234013)	Relay changes by fewer than 10 points in every cell	7 of 12 cells changed more (six rose by 11–43 points, one fell by 23)	failed
NC2	No-concerns prompt	Unprompted doubt falls below half of solicited flags	6 of 12 cells	partly held
KW	No-concerns prompt	The keyword detector agrees with hand labels on 50 runs	$\kappa = 0.39$ (13 of 20 flags missed); replaced by a classifier ($\kappa = 0.84$)	failed
FD1	Familiar field (9234013)	A familiar copy raises relay over the unfamiliar copy by at least 10 points, pooled	+9 points ($p = 0.015$); Opus 5 +13, Argon +15	failed
FD2	Familiar field	A familiar copy does not raise relay over the weight alone by 10 points	149 against 148 of 234	held
FD3	Familiar field	A familiar disagreeing copy lowers relay	21 against 149 of 236	held
CL	Superseded, addendum (cfbf0a7)	Opus 5 clean accuracy on the 79 superseded-record tasks is at least 95%	79 of 79	held

Supplementary Table 22: Held-out validation of the unprompted-doubt classifier: 100 runs not used in its development (60 without the concerns field, 40 under the registered prompt), labelled by an independent annotator blind to the classifier, whose answers were committed before labelling (commit 6964e2e). Cohen's κ 0.90 (95% bootstrap interval 0.80–0.98).

Prompt	Runs	Agree	Classifier only	Annotator only	κ	Annotator flags	Classifier flags
no concerns	60	56	4	0	0.87	27/60	31/60
registered	40	39	1	0	0.93	30/40	31/40
all	100	95	5	0	0.90	57/100	62/100